

Comparing the Different Operating Systems

Features	iOS	Android	Firefox OS	Windows Phone	Blackberry OS	Ubuntu Touch
Company	Apple Inc.	Open handset alliance/Google	Mozilla foundation	Microsoft	Blackberry Ltd	Canonical Ltd
OS family	Darwin	Linux	Linux	Windows NT 8 +	QNX	Linux
Supported CPU architecture	ARM and ARM64	ARM, x86 MIPS and 64-bit variants of all three	ARM and x86	ARM	ARM	ARM and x86
Programming language	C, C++, Objective C and Swift	C, C++ and Java	HTML5, CSS, JavaScript and C++	.NET C#, VB.NET, Silverlight, native C/C++, WinRTP (XMLA) and DirectX	C, C++, Native SDK and HTML5/ Javascript/CSS	C, C++ and QML
Package manager	iTunes	APK	Firefox OS packaged apps	Zune software	Blackberry link	Click packages and dpkg
Remote device location tracking	Yes	Yes	No	Yes	Yes	-
Remote device locking and/or data wipe	Yes	Yes	No	Yes	Yes	-
Accessibility features	Yes	Yes	No	Yes	Yes	-
On-device encryption	Yes	Yes	No	Yes	Yes	-
Sync to cloud encryption	Yes	Yes	-	Yes	Yes	-
Desktop-sync	Yes	No	No	Yes	Yes	-
Major available browsers	Safari and Chrome	Chrome and Firefox	Firefox	IE and Opera Mini	-	-
Official application store	App Store	Google Play	Firefox Market-place	Windows Phone Store	Blackberry World	Ubuntu Store
Printer support	Yes(AirPrint)	Yes; over cloud print	-	Third-party software	Yes	-
Multitasking	Yes	Yes	Yes	Yes	Yes	Yes
Bluetooth keyboard	Yes	Yes	-	No	Yes	-
VPN	Yes	Yes	No	Yes	Yes	-

vices and libraries. This architecture can, perhaps, best be represented visually as in Fig. 9. Each layer of the stack and the corresponding elements within each layer are tightly integrated and carefully tuned to provide an optimal application development and execution environment for mobile devices.

Linux Kernel. The basic layer is Linux Kernel. The whole Android OS is built on top of Linux Kernel with some further architectural changes.

Linux Kernel interacts with the hardware and contains all essential hardware drivers. It also acts as an abstraction layer between the hardware and other software layers. The kernel provides pre-emptive multitasking, low-level core system services such as memory, process

and power management, in addition to providing a network stack and device drivers for hardware such as the device display, Wi-Fi and audio.

Libraries. The next layer is Android's native libraries. It is this layer that enables the device to handle different types of data. These libraries are written in C or C++ language and are specific to a particular hardware.

Some of the important native libraries include the following:

Surface Manager. It is used for compositing window manager with off-screen buffering. Off-screen buffering means that the apps cannot directly be drawn into the screen, instead drawings go to the off-screen buffer. There, it is combined with other drawings and form the final

screen that the user will see. This off-screen buffer is the reason behind the transparency of windows.

Media framework. Media framework provides different media codecs allowing the recording and playback of different media formats.

SQLite. SQLite is the database engine used in Android for data-storage purposes.

WebKit. It is the browser engine used to display HTML content.

OpenGL. This is used to render 2D or 3D graphics content to the screen.

Android Runtime. Android Runtime consists of Dalvik virtual machine and Core Java libraries.

Dalvik virtual machine. It is a type of JVM used in Android devices to run apps, and is optimised for low